# Susceptibility of Stem Cells to Deleterious Mutations

According to the U.S. National Library of Medicine, there are currently more than 2,300 ongoing clinical trials around the world that involve stem cells. As researchers, now more than ever, look to stem cells for prospective therapies, there is growing concern about the safety of using these regenerative cells. Specifically, in recent years, researchers have increased efforts to study the effects of mutations that are derived in vivo and in vitro. While pharmaceutical companies have heralded stem cells as versatile (and relatively uncomplicated) means of delivering potential treatments for complex diseases like cancer, diabetes, or Parkinson's disease, these cells remain susceptible to deleterious mutations, contributing to why the safety of stem cell therapies is questionable.

As described in a study by Cannataro et al., stem cells exist in specifically-sized populations throughout the human body. For example, in the small intestines, a particular niche (microenvironment that harbors stem cells and other cells related to stem cell regulation) contains about twenty stem cells. Furthermore, the number of stem cells per niche varies throughout the human body as well as from organism to organism (e.g., while the niches of the small intestines in humans contain about twenty stem cells, the niches of the small intestines in mice contain about seven stem cells). The reasoning behind the specific number of stem cells per niche is due primarily to a single evolutionary trade-off: altering the number of stem cells per niche adversely affects the probability that one or more of those cells will become tumorigenic; however, at the set number of stem cells per niche, the stem cells are at greater risk (compared to niches with different numbers of stem cells) of developing deleterious mutations that cause aging and tissue degradation (Cannataro et al. 2017).

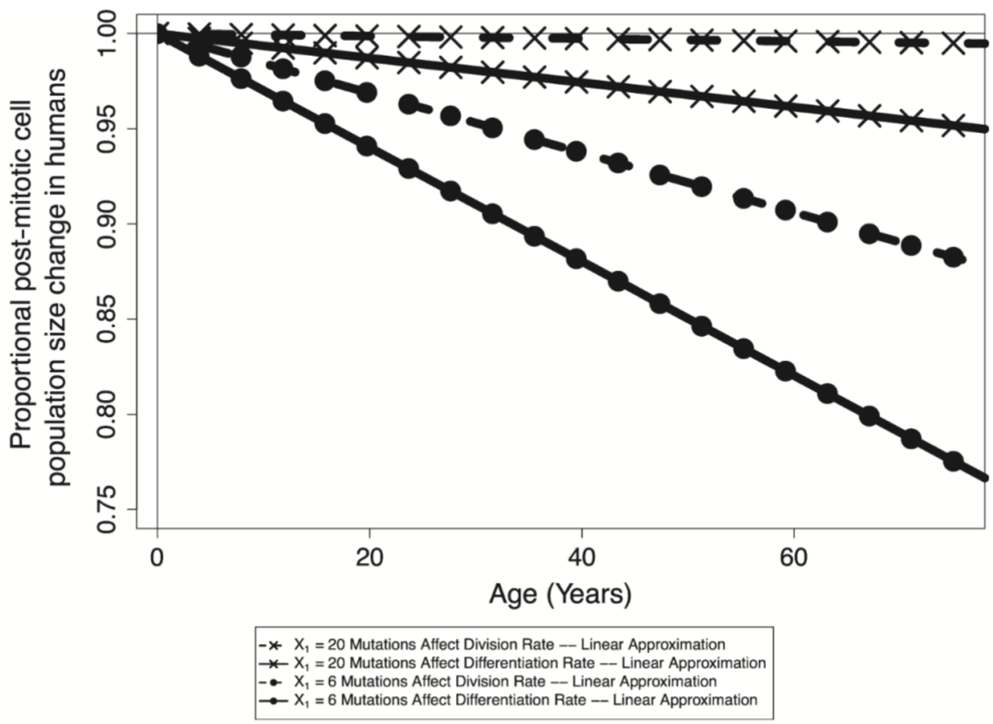


Fig. 1. The amount of human small intestine tissue changes over time as mutations accumulate; small intestine tissue containing niches of twenty stem cells will retain far more of its mass than small intestine tissue containing niches of only six stem cells (Cannataro et al.).

Though, organisms with a lower risk for tumorigenesis (at the expense of a greater risk for rapid aging and tissue degradation) are more evolutionarily fit since tumorigenesis is more likely than aging or tissue degradation to harm survival and reproductive ability within a short period of time. Moreover, human small intestine tissue with twenty stem cells per niche retains cells to a far greater degree than small intestine tissue with only six stem cells per niche (e.g., see Fig. 1), leading to the inference that mutations causing tumorigenesis have greater deleterious effects on tissue loss than do mutations causing aging and tissue degradation (Cannataro et al.). And since the evolutionary trade-off regarding stem cells is not unique to stem cells of the human small intestine, the study by Cannataro et al. also provides insights for stem cells throughout the rest of the human body and in other organisms.

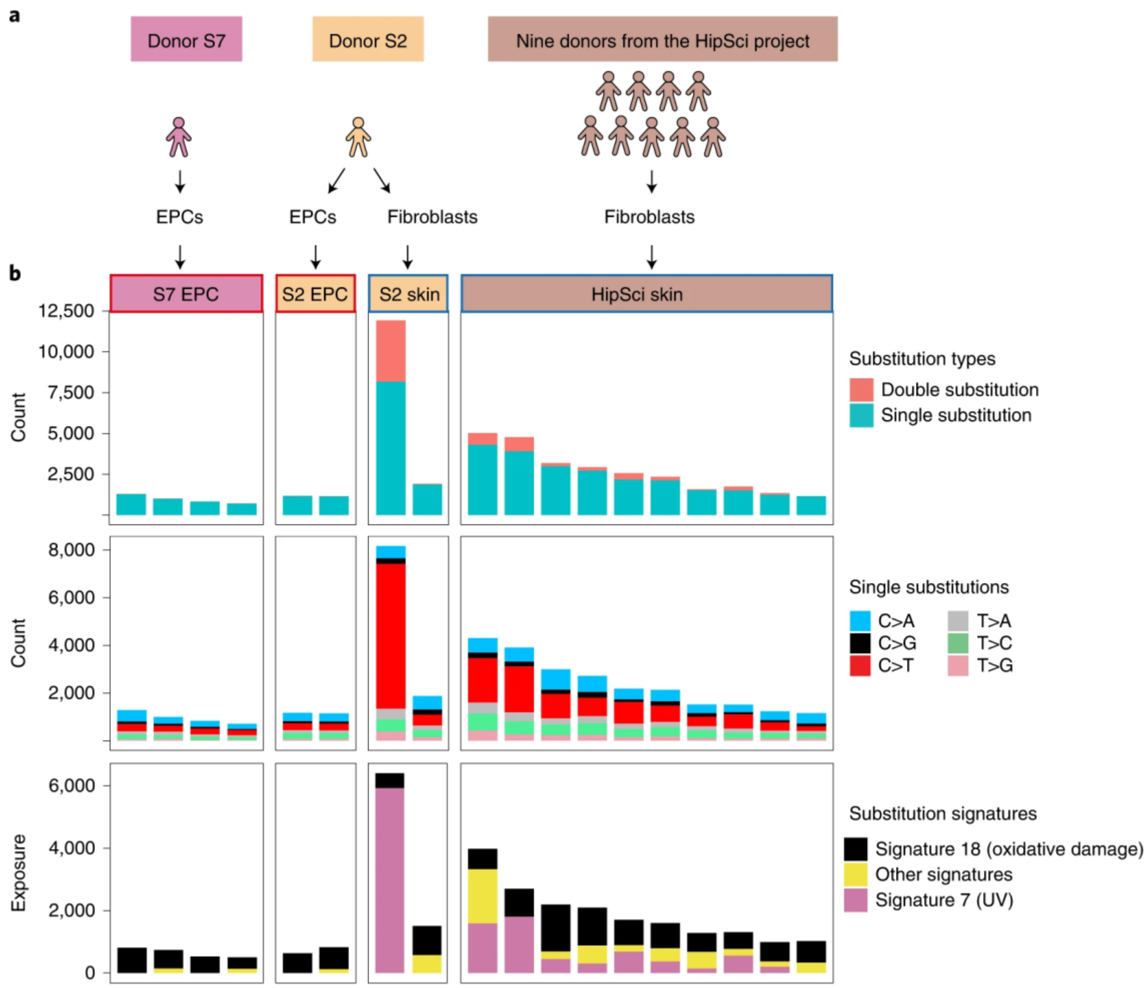


Fig. 2. The bar graphs display relative amounts of single-base pair substitutions in endothelial progenitor cells (progeny of blood-derived stem cells; columns 1 and 2) and fibroblasts (progeny of skin-derived stem cells; columns 3 and 4), along with their respective substitution types (Rouhani et al).

Other deleterious mutations in stem cells occur in vivo because of exposure to UV light. A study by Rouhani et al. explored the genetic variations between stem cells derived from skin and blood, finding that skin-derived stem cells had significantly greater mutational burdens than blood-derived stem cells (e.g., see Fig. 2). Not only did more single-base pair substitutions occur in skin-derived stem cells but UV-related substitutions occurred *only* in skin-derived stem cells. As indicated by signature 7, UV-related substitutions accounted for about half of all single-base pair substitutions in skin-derived stem cells (e.g., see Fig. 2). According to the study data, 72% of all tested skin-derived stem cells exhibited some degree of substitution signature 7, and there was considerable heterogeneity (cells with different mutational burdens produced the same phenotypic effect) among these cells. The heterogeneity among these stem cells is most likely related to their differing levels of sunlight exposure; certain stem cells had suffered longer exposure to UV light from the sun and, thereby, accumulated more UV-related mutations

(Rouhani et al. 2022). Because of the asexual nature by which mitosis occurs, UV damage to stem cells affects future generations of skin-derived stem cells as well as the specialized cells which these stem cells become.

Another large proportion of single-base pair substitutions in stem cells occurs because of oxidative stress in vitro; however, this type of single-base pair substitution can affect all types of stem cells, not only skin-derived stem cells. In a study by Kuijk et al., researchers acquired liver, intestinal, and pluripotent (found in bone marrow, blood, and adipose fat tissue) stem cells from human patients in order to determine the mutational burdens of each stem cell type. Their findings indicated a high prevalence of C>A single-base pair substitutions, accounting for at least one-third of all single-base pair substitutions in these stem cell types (Kuijk et al).

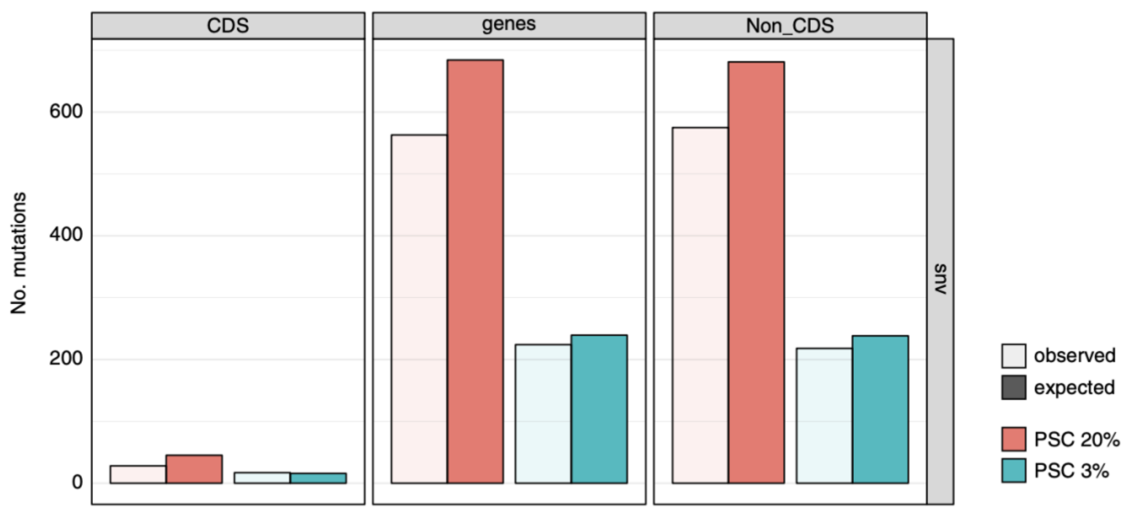


Fig. 3. The bar graph displays relative observed and expected values for the number of oncogenic mutations (mutations that occur on genes linked to cancer) under normal and decreased oxygen presence (Kuijk et al.).

Researchers have previously linked the C>A substitution to increased prevalence of reactive oxygen species (which cause oxidative stress). Compared to uncultured stem cells from the same patients, the different stem cell types used in this study contained, on average, almost forty times as many single-base pair substitutions, proving that oxidative stress is far more prevalent in vitro than in vivo. Under typical atmospheric conditions (an $O_2$ concentration of 20%), the number of oncogenic mutations is well over double the number of oncogenic mutations that occur at an $O_2$ concentration of 3% (Kuijk et al.). The change in the number of oncogenic mutations by decreasing atmospheric oxygen concentration suggests that a large proportion of in vitro mutations occur due to the fact that stem cells experience greater exposure to oxygen while outside of the human body.

All stem cells carry deleterious mutations that warrant concern by researchers who seek to create stem- cell-derived therapies. It is important to understand the probable mutational burden that is characteristic of stem cells as well as the increase in mutational burden that occurs during in vitro culturing. Researchers can certainly minimize mutational burden by not selecting skin-derived stem cells for therapies as well as by controlling the oxygen content of the air that is present during culturing. Beyond that, the minimization of deleterious mutations in stem cells is likely uncontrollable; however, since all stem cells are subject to mutations that cause aging and cell degradation, lack of further mutation prevention may not be necessary to yield a level of mutational burden that is roughly natural.

## Literature Cited

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